Extensive vs. Intensive Accumulation: Choice of Technique and Capacity Utilization

Preliminary CPR Estimation Results for the US Corporate Sector

Dissertation Chapter 2

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Bottom Line Up Front (BLUF)

- **Core Question:** How do distributive struggles and capital structure condition long-run productive capacity?
- **Methodology:** We resolve the $I(2)$ cointegration trap by treating the levels relationship as a **Cointegrating Polynomial Regression (CPR)** estimated via FM-OLS and IM-OLS.
- **Multicollinearity Solution:** Residual centering (orthogonalization) stabilizes the design matrix, reducing interaction VIFs from > 80.0 to exactly **1.0**.
- **Key Empirical Result:**
  - We find a **negative and highly significant** interaction between intensive accumulation (τ) and the wage share (ω) in the Fordist era, validating the induced innovation theory on a concave frontier.
  - A stark **structural break** occurs post-1974: the interaction becomes positive and statistically non-significant, reflecting the rise of financialization and global supply chains.

The Extensive Margin (K^{NRC})

Ontological Definition

Nonresidential structures and infrastructure (K^{NRC}) represent the physical plant capacity, logistical networks, and spatial layout of production.

- **Temporal Properties:** High lumpiness, long gestation periods, and high irreversibility.
- **Decision Rule:** Evolving along a **normal reproduction corridor** governed by credit and institutional closure rather than marginal optimization:

$$g^N_{K^{NRC},t} = \chi^N r^N_t - \delta$$

- **Independence from Conflict:** $\frac{\partial K^{NRC}}{\partial \omega} = 0$. Multi-year planning horizons filter out short-run distributive shocks (ω).
- **Analytical Role:** Infrastructure acts as a structural envelope, strictly conditioning the parameters ($\lambda^{[s]}_i$) of the technological frontier.

The Intensive Margin (K^{ME}/L) & Technique Choice

Ontological Definition

Machinery (K^{ME}) paired with labor (L) defines the tactical margin of technique choice ($q = K^{ME}/L$).

- **Tactical Cost-Minimization Problem:**

$$\min_q C = a - \pi q \quad \text{s.t.} \quad a = \lambda_0^{[s]} + \lambda_1^{[s]} q + \lambda_2^{[s]} q^2$$

  where a is labor productivity growth, and $\pi = 1 - \omega$ (profit share).

- **First-Order Condition (FOC):**

$$\frac{\partial C}{\partial q} = \lambda_1^{[s]} + 2\lambda_2^{[s]} q^* - \pi = 0 \implies q^*(\pi, s) = \frac{\pi - \lambda_1^{[s]}}{2\lambda_2^{[s]}}$$

- **Induced Innovation:** Since the frontier is concave ($\lambda_2^{[s]} < 0$):

$$\frac{\partial q^*}{\partial \omega} = -\frac{1}{2\lambda_2^{[s]}} > 0 \quad (\text{since } \omega \uparrow \implies \pi \downarrow)$$

  Rising wage share induces faster mechanization ($q^* \uparrow$) to save labor.

Reconciliation: Diminishing Capacity Payoffs

- Differentiating $Y_t^p = \frac{a^N}{q^*} K_t^{ME} + \Gamma(K_t^{NRC})$ yields transformation elasticity θ_t:

$$\theta_t \approx \theta^{ME} = \frac{a^N}{q^*}$$

- Substituting the concave frontier ($a^N = \lambda_0 + \lambda_1 q^* + \lambda_2 (q^*)^2$):

$$\theta^{ME} = \frac{\lambda_0}{q*} + \lambda_1 + \lambda_2 q^*$$

- Differentiating with respect to mechanization rate q^*:

$$\frac{\partial \theta^{ME}}{\partial q^*} = \frac{\lambda_2 (q^*)^2 - \lambda_0}{(q^*)^2} < 0 \quad \text{(since } \lambda_2 < 0 \text{ and } \lambda_0 > 0\text{)}$$

- **The Distributive Link:** Distributive pressure ($\omega \uparrow \implies q^* \uparrow$) drives the system into lower marginal capacity conversion efficiency:

$$\frac{\partial \theta^{ME}}{\partial \omega} = \frac{\partial \theta^{ME}}{\partial q^*} \frac{\partial q^*}{\partial \omega} < 0$$

- **Verdict:** The interaction coefficient between composition (τ) and wage share (ω) must be **negative**.

Empirical Specifications

- **Specification A (Direct Scale-Conditioning):**

$$y_t = \alpha + \theta_0 \tilde{k}_t^{scale} + \phi\tilde{\omega}_t + \gamma(\tilde{k}_t^{scale}\tilde{\omega}_t) + \varepsilon_t$$

  where k^{scale} is plant scale (k_{NRC}) or total corporate capital (k_{Kcap}).

- **Specification B (Composition-Mediated Conditioning):**

$$y_t = \alpha + \theta\tilde{k}_t^{scale} + \psi\tilde{\tau}_t + \phi\tilde{\omega}_t + \lambda(\tilde{\tau}_t\tilde{\omega}_t) + \varepsilon_t$$

  where $\tau = k_{ME} - k_{NRC}$ isolates the physical machinery-to-structures ratio.

- **The Econometric Resolution:**
  - Regressors are mixed-order integrated: the interaction term behaves as $I(2)$ / $I(2)$ risk. We use **IM-OLS** and **FM-OLS** to obtain superconsistent, asymptotically valid inferences.
  - **Residual Centering:** Regressing $(\tilde{\tau}_t\tilde{\omega}_t)$ on main effects extracts the orthogonalized interaction, reducing VIFs to exactly 1.0.
  - **Ex-Post Robustness Check:** Both Engle-Granger (EG) residual unit root tests and Phillips-Ouliaris (PO) multivariate cointegration tests verify the stationary cointegrating vector.

Results: Specification A (Scale-Conditioning)

Table: Specification A: Direct Scale-Conditioning via K^{NRC} (FM-OLS, Orthogonalized)

Window	Regressor	Coeff	t-stat	p-value
Full Sample (1929–2024)	$k_{NRC,centered}$	1.0228	17.6021	0.0000
	$\omega_{NFC,centered}$	-12.4011	-6.3539	0.0000
	Interaction (orth)	-7.3445	-3.1766	0.0020
Fordist Core (1929–1973)	$k_{NRC,centered}$	0.8387	28.4347	0.0000
	$\omega_{NFC,centered}$	0.7624	0.2655	0.7921
	Interaction (orth)	-1.1549	-0.6827	0.4988
Post-Fordist (1974–2024)	$k_{NRC,centered}$	1.0916	6.3413	0.0000
	$\omega_{NFC,centered}$	-23.4366	-9.2208	0.0000
	Interaction (orth)	28.6878	5.0074	0.0000

- Scale coefficient is positive and robustly close to 1.0.
- **Robustness:** Cointegration confirmed; residuals reject the null of no cointegration under both Engle-Granger and Phillips-Ouliaris tests.

Results: Specification B (Composition-Mediated)

Table: Specification B: Composition-Mediated (FM-OLS, Orthogonalized)

Window	**Regressor**	**Coeff**	**t-stat**	**p-value**
Golden Age (1945–1973)	$k_{NRC,centered}$	0.2378	3.7248	0.0010
	$\tau_{centered}$	0.3121	10.5266	0.0000
	$\omega_{NFC,centered}$	-5.1605	-3.6241	0.0014
	Interaction (orth)	**-1.0294**	**-2.4892**	**0.0201**
Post-War Fordist (1940–1973)	$k_{NRC,centered}$	0.0773	1.1122	0.2752
	$\tau_{centered}$	0.3809	11.7582	0.0000
	$\omega_{NFC,centered}$	-7.6639	-4.8891	0.0000
	Interaction (orth)	**-2.0342**	**-4.9555**	**0.0000**
Post-Fordist (1974–2024)	$k_{NRC,centered}$	0.2146	4.2083	0.0001
	$\tau_{centered}$	0.3520	16.9510	0.0000
	$\omega_{NFC,centered}$	1.1260	0.7747	0.4425
	Interaction (orth)	0.0821	0.1257	0.9006

- Composition effect (τ) is highly significant. Interaction is negative and significant during the Fordist era.
- **Robustness:** Cointegration validated by both Engle-Granger and Phillips-Ouliaris tests, ensuring stationary residuals.

Critical Discussion: Validation vs. Structural Crisis

Validation of the Theory

The negative and highly significant interaction coefficient ($\lambda < 0$) in the Fordist era ($1940 - 1973$) strongly validates the induced innovation theory. Higher wage shares (ω) force capitalists to mechanize, but this drives the economy into the concave zone of lower marginal capacity conversion.

The Post-Fordist Structural Break

Post-1974, the interaction collapses to zero (0.0821, p-value 0.90).

- **Global Offshoring:** Domestic wage bargaining no longer drives technique choice in the same way because capital can relocate.
- **Financialization:** Capital accumulation shifts away from real plant investment to financial channels.
- **Off-shoring/Subcontracting:** The domestic wage share ω_t ceases to act as a reliable proxy for the total cost of labor involved in global production chains.

Conclusion & Next Steps

- We have established a robust, cointegrated long-run relationship using Cointegrating Polynomial Regression (CPR) and residual centering.
- The results confirm that capital composition (τ) mediates capacity conversion, and that this mediation is conditioned by distribution (ω) in the Fordist core.
- **Next Steps (Stage S40):**
  - Reconstruct the potential capacity output series:

$$\hat{y}_t^p = \hat{\alpha} + \hat{\theta}\tilde{k}_t^{scale} + \hat{\psi}\tilde{\tau}_t + \hat{\phi}\tilde{\omega}_t + \hat{\lambda}(\tilde{\tau}_t\tilde{\omega}_t)_{orth}$$

  - Recover the latent capacity utilization series as the stationary cointegrating residual:

$$\widehat{\ln \mu_t} = y_t - \hat{y}_t^p$$